

34721/34722 Linear Control Design 1

Spring Semester



Teachers: Silvia Tolu - Dimitrios Papageorgiou

Lesson 4/13: Frequency Domain

Schedule

13:00-15:00

- **Properties in the frequency domain**
- Poles and zeros in the s-plane
- S-plane, pole, and zero-point location
- Time analysis
- 2nd order systems

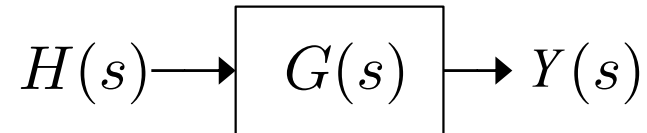
15:00 - 17:00

- Intro to Regbot

Day	Topics	Contents	Assignments
1	Introduction	Intro to feedback control, Pre-Test	MATLAB and Simulink
2	Control concepts	Block diagram, control process, hand tuning	Hand tuning, Ziegler-Nichols
3	Transfer function and Laplace	Laplace and transfer function	Block diagrams and transfer functions
→ 4	Frequency domain	Frequency domain properties	Introduction to REGBOT
5	Modelling and Linearization	White/black box modelling and linearization	REGBOT – Transfer function from data
6	Bode Plots and stability	Bode plot, stability margins, P-design	REGBOT – Bode, stability, P-Controller design

Properties in the frequency domain

Steady-State (or Static) Gain



$$H(s) = \frac{h_0}{s} \quad h_0 = \text{DC value of input}$$

The **steady-state (or static) gain** of the **TF**:

$$K_{ss} = \lim_{s \rightarrow 0} G(s)$$

It applies only to stable systems

$$s = \sigma + j\omega$$

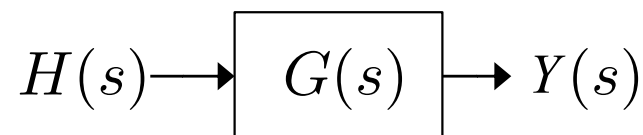
ω Signal “real” frequency
 σ “Imaginary” frequency;
zero for real frequencies

$$f(t) = \begin{cases} 0 & t < 0 \\ f(t) & t > 0 \end{cases}$$

The **steady-state gain** is also called static gain or the **DC gain**, as it corresponds to the gain at zero frequency

Properties in the frequency domain

Steady-State (or Static) Gain



$$H(s) = \frac{h_0}{s}$$

$$G(s) = \frac{b}{s+a}$$

$$G(s) = \frac{b}{a} \frac{a}{s+a}$$

$$K_{ss} = \lim_{s \rightarrow 0} G(s) = \frac{b}{a}$$

Steady-state Output

$$Y(s) = \frac{h_0}{s} \frac{b}{a} \frac{a}{s+a}$$

$$\frac{a}{s(s+a)} = \frac{k_1}{s} + \frac{k_2}{s+a}$$

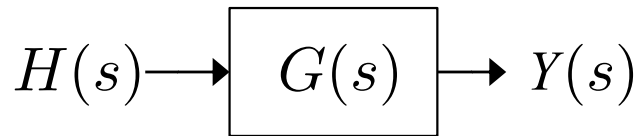
Fraction expansion

$$\frac{s a}{s(s+a)} = \frac{s k_1}{s} + \frac{s k_2}{s+a} \Big|_{s=0} \longrightarrow k_1 = 1$$

$$\frac{a(s+a)}{s(s+a)} = \frac{(s+a) k_1}{s} + \frac{(s+a) k_2}{s+a} \Big|_{s=-a} \longrightarrow k_2 = -1$$

Properties in the frequency domain

Steady-state (or static) Gain



$$H(s) = \frac{h_0}{s}$$

$$G(s) = \frac{b}{s + a}$$

$$G(s) = \frac{b}{a} \frac{a}{s + a}$$

$$K_{ss} = \lim_{s \rightarrow 0} G(s) = \frac{b}{a}$$

Steady-state Output (Y_{ss})

$$Y(s) = \frac{h_0}{s} \frac{b}{a} \frac{a}{s + a}$$

$$Y(s) = h_0 \frac{b}{a} \left(\frac{1}{s} - \frac{1}{s+a} \right)$$

Inverse Laplace

$$y(t) = h_0 \frac{b}{a} (1 - e^{-at})$$

$f(t)$	$F(s)$
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s + a}$
$Af(t)$	$AF(s)$

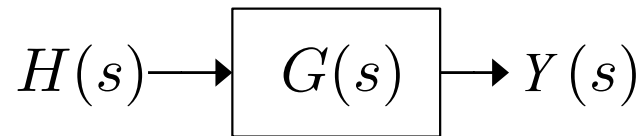
Final value theorem

$$Y_{ss} = \lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s Y(s)$$

$$Y_{ss} = h_0 \frac{b}{a} 1 = h_0 \frac{b}{a}$$

$$\lim_{t \rightarrow \infty} (1 - e^{-at}) = 1$$

Example



$$K_{ss} = \lim_{s \rightarrow 0} G(s)$$

$$Y_{ss} = \lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s H(s) G(s)$$

$$H(s) = \frac{1}{s} \quad G(s) = \frac{220}{s^2 + 10s + 100}$$

Steady-State Gain $K_{ss} = \lim_{s \rightarrow 0} G(s) = \frac{220}{100} = 2.2$

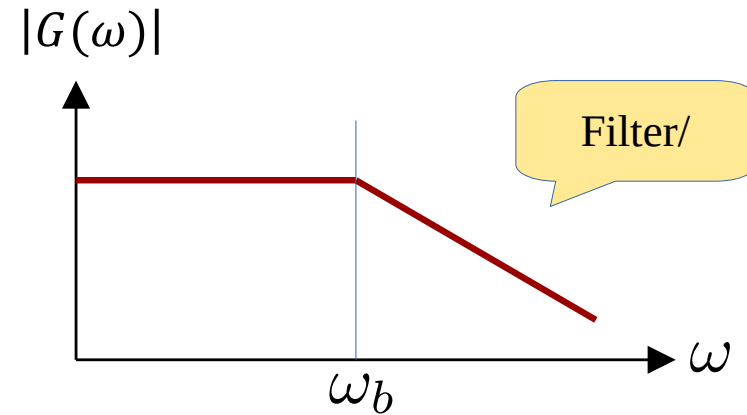
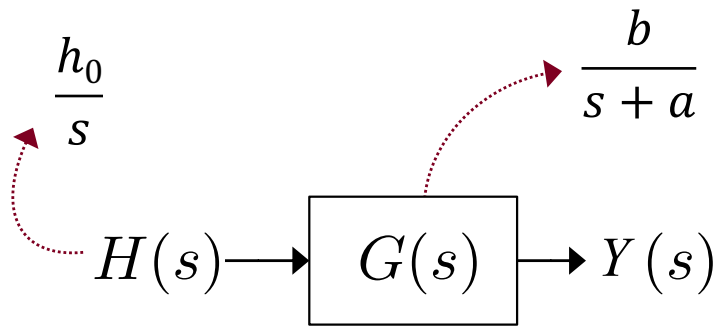
Steady-State Output $Y_{ss} = \lim_{s \rightarrow 0} s \frac{1}{s} G(s) = 2.2$

In Matlab → function: *dcgain*

Properties in the frequency domain

Break frequency - Time constant

$$s = \sigma + j\omega$$



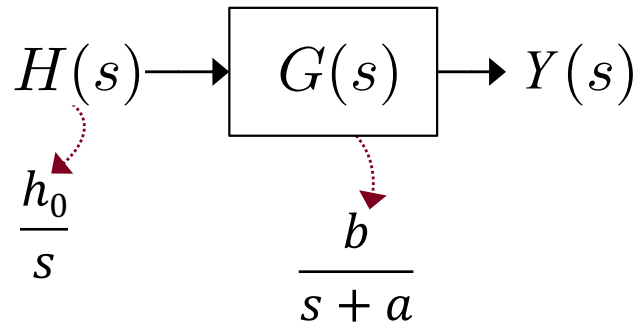
$$G(s) = K_{ss} \frac{\omega_b}{s + \omega_b} \quad \omega_b = \text{Break frequency [rad/sec]}$$

$$G(s) = K_{ss} \frac{1}{\frac{1}{\omega_b} s + 1}$$

$$G(s) = K_{ss} \frac{1}{\tau_b s + 1} \quad \frac{1}{\omega_b} = \tau_b = \text{time constant [sec]}$$

Properties of the frequency domain

Time Constant τ_b



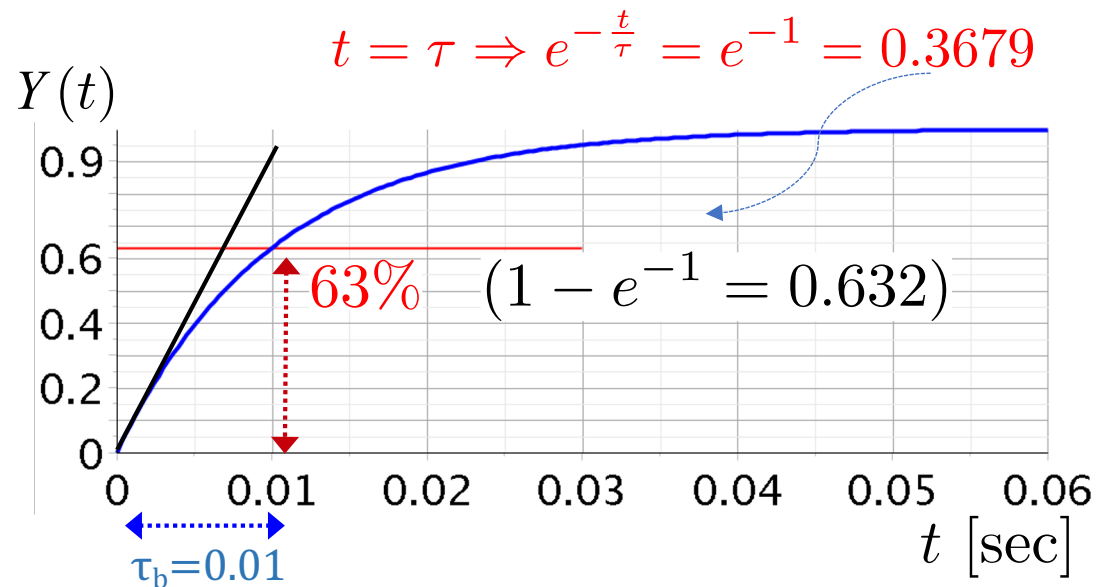
$$Y(s) = \frac{h_0}{s} K_{ss} \frac{\omega_b}{s + \omega_b}$$

$$Y(t) = h_0 K_{ss} (1 - e^{-\omega_b t})$$

$$e^{-\omega_b t} = e^{-\frac{t}{\tau_b}} \Rightarrow \text{time constant } \tau_b = \frac{1}{\omega_b}$$

τ_b expresses the time the step or impulse response takes to reach 63.2% of the way from a start value to the *steady-state* value

The tangent intersects the steady-state value when $t = \tau_b$



Unit step response $h_0 = 1, K_{ss} = 1, \omega_b = 100$

Control Questions

For the transfer function

$$G(s) = \frac{150}{3s + 30}$$

Calculate:

- a) The steady-state gain K_{ss} of $G(s)$
- b) The break frequency ω_b

a) The steady-state gain

$$G(s) = \frac{b}{a} \frac{a}{s + a} = \frac{150}{3} \frac{1}{10} \frac{10}{s + 10} \rightarrow K_{ss} = \lim_{s \rightarrow 0} \frac{150}{3s + 30} = 5$$

b) The break frequency

$$G(s) = K_{ss} \frac{\omega_b}{s + \omega_b}$$

$$G(s) = \frac{b}{a} \frac{a}{s + a} = \frac{150}{30} \frac{10}{s + 10}$$



$$\omega_b = 10 \text{ [rad/sec]}$$

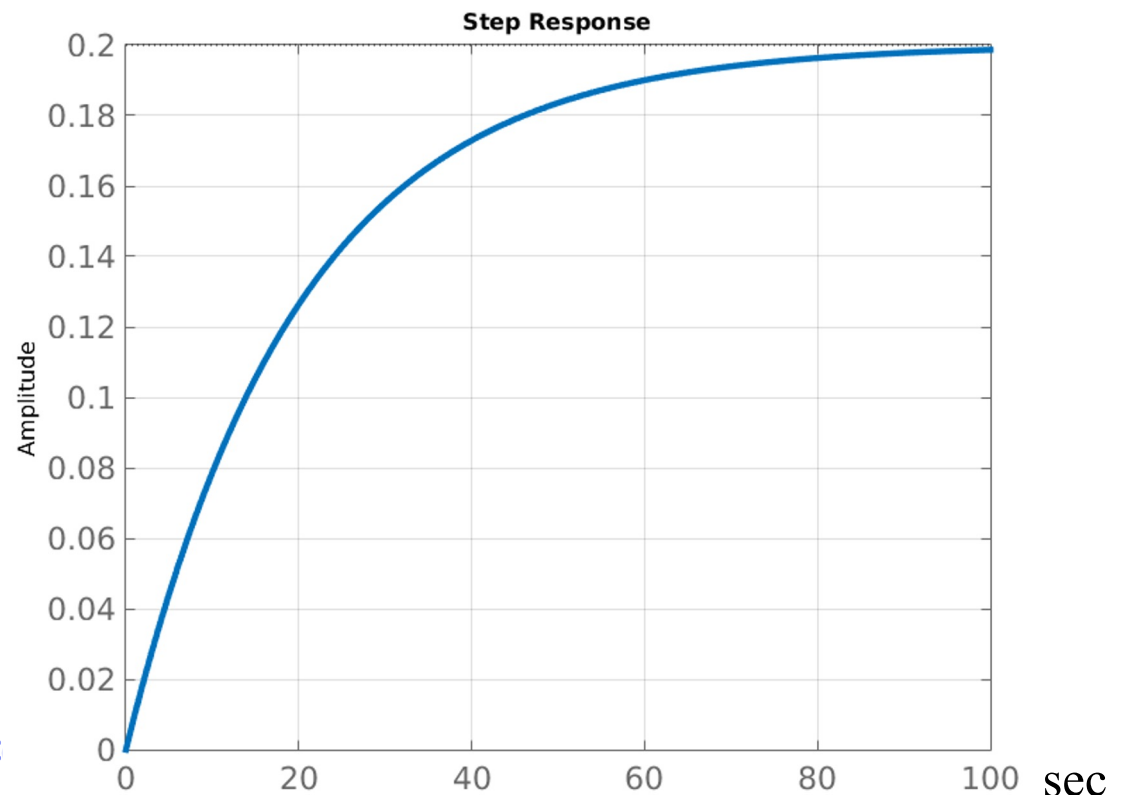
Control Questions



The plot shows the output of a unit step input to a transfer function which is assumed to have only one zero at the denominator (pole).

Calculate:

- a) The steady-state gain K_{ss}
- b) The break frequency ω_b
- c) The transfer function $G(s)$



Control Questions



2) The plot to the right shows the output of a unit **step input** to a transfer function that is assumed to have only one zero at the denominator (**pole**).

$$h_0 = 1$$

a) **Steady-state gain**

$$Y(t) = h_0 K_{ss} (1 - e^{-\omega_b t})$$

$$K_{ss} = \frac{y_{ss}}{h_0} = 0.2$$

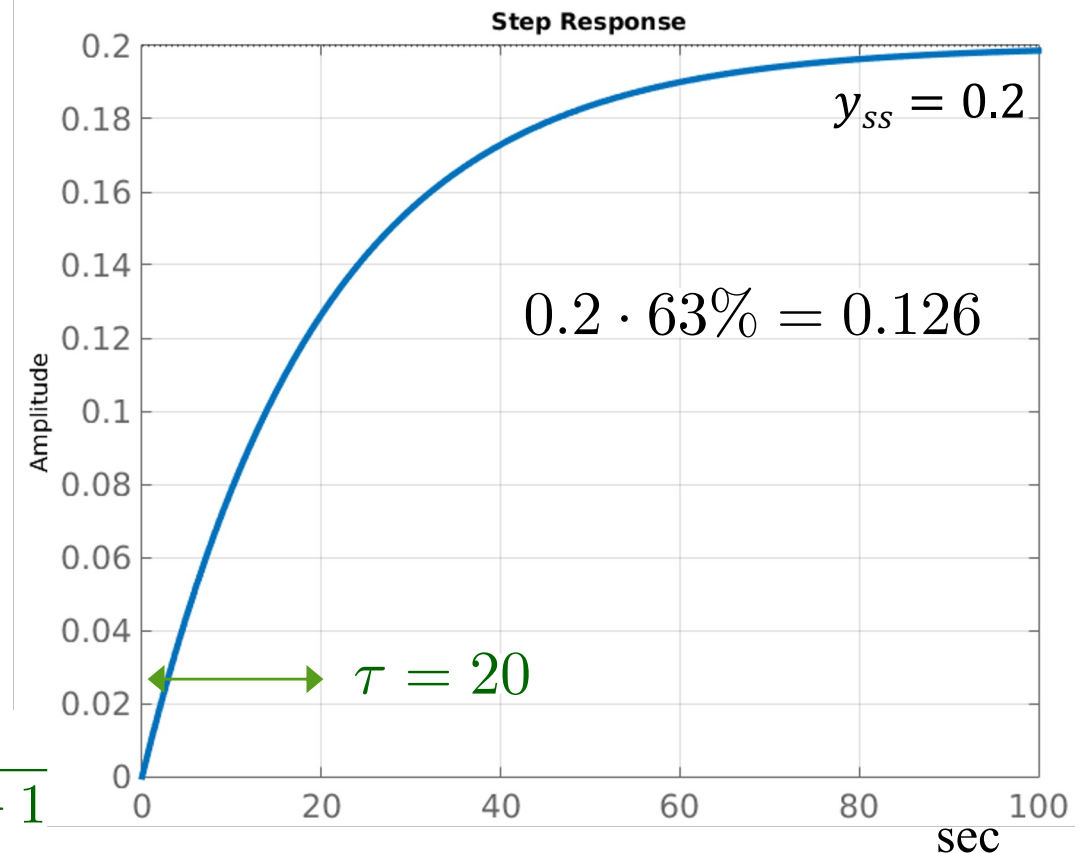
b) **Break frequency**

$$\omega_b = \frac{1}{\tau} \Rightarrow \omega_b = 0.05 \frac{\text{rad}}{\text{s}}$$

c) **Transfer function**

$$G(s) = 0.2 \frac{0.05}{s + 0.05}$$

$$G(s) = K_{ss} \frac{1}{\tau s + 1} \Rightarrow G(s) = 0.2 \frac{1}{20s + 1}$$



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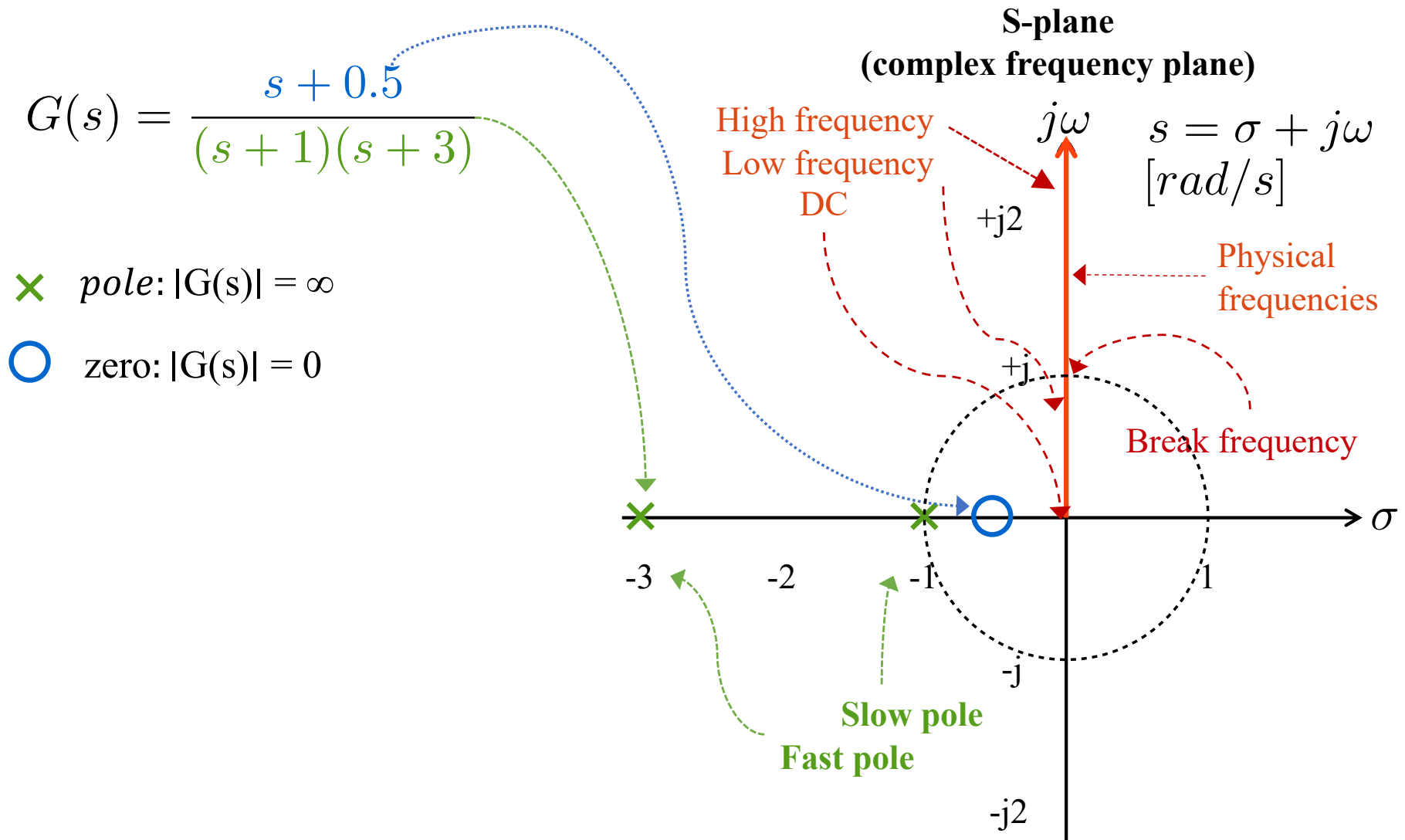
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Poles and Zeros (S-plane)



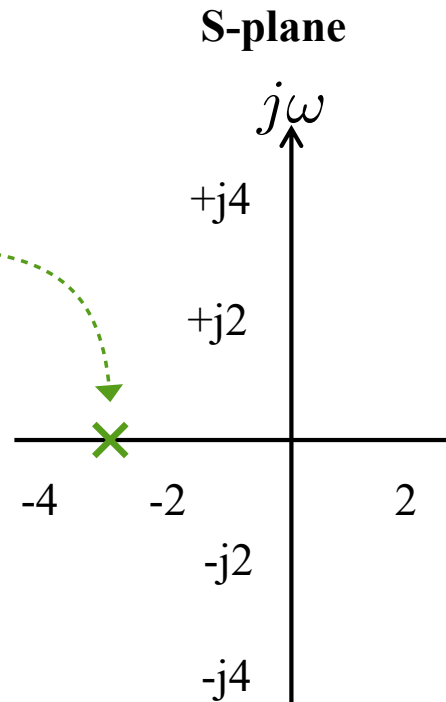
Poles (1st order): what is the consequence for y(t)?



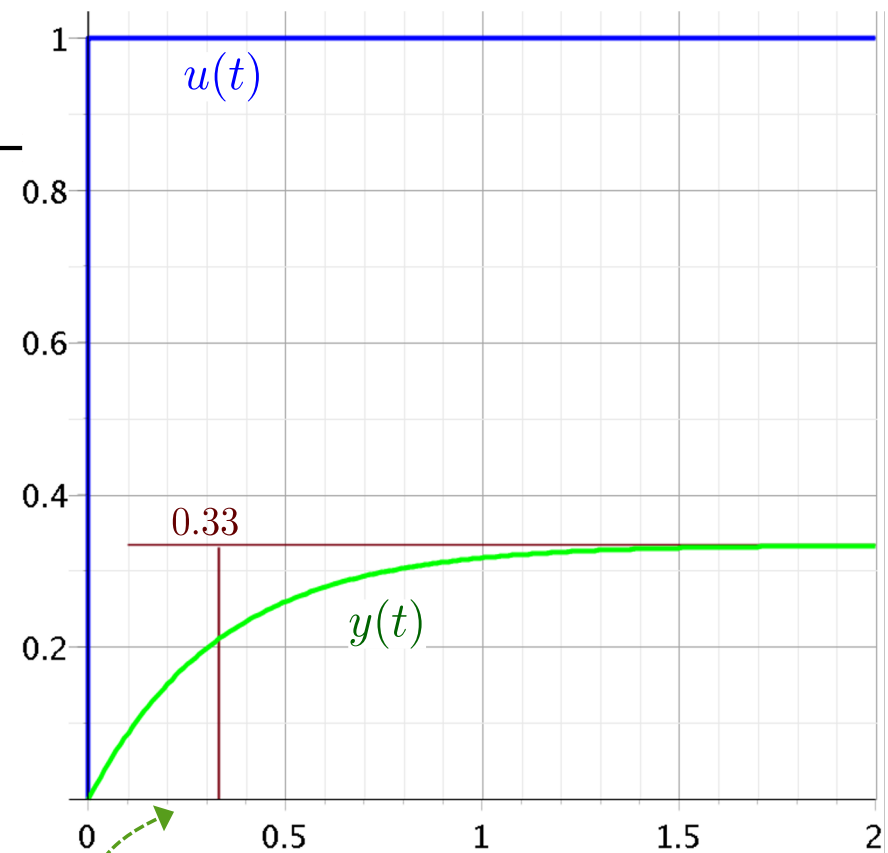
(in the left half-plane)

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s + 3}$$

$$Y(s) = U(s) \frac{1}{s + 3}$$



$$U(s) = \text{step} = \frac{1}{s}$$



$$G(s) = K S_s \frac{\omega_b}{s + \omega_b} = K S_s \frac{1}{\tau_b s + 1}$$

Break Frequency

$$\tau_b = \frac{1}{\omega_b}$$

Time Constant

$$\tau = \frac{1}{3}$$

Poles (1st order): what is the consequence for y(t)?



(in the left half-plane)

S-plane

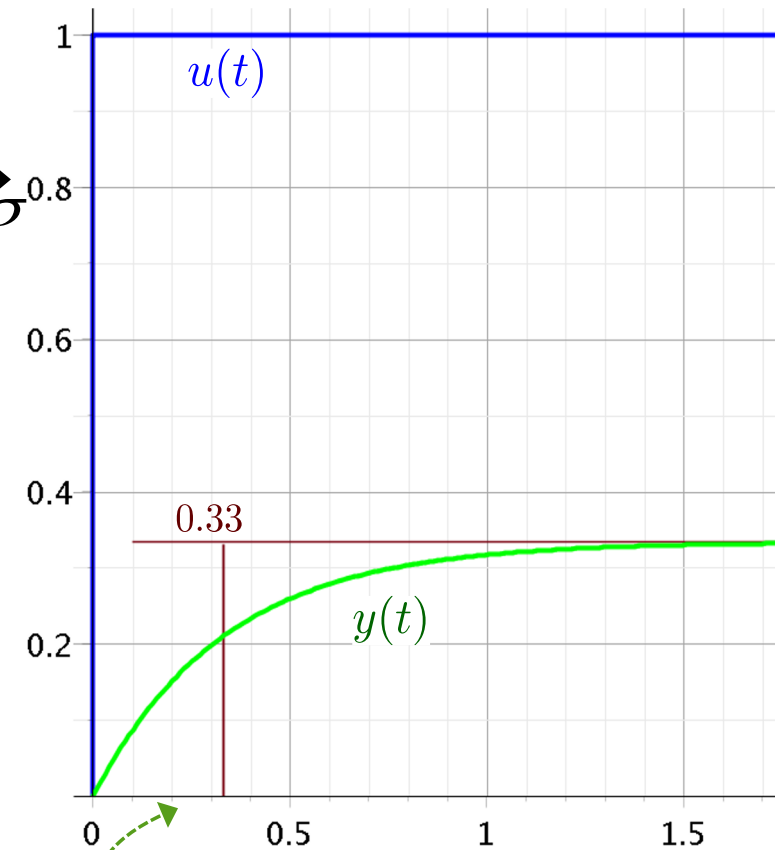
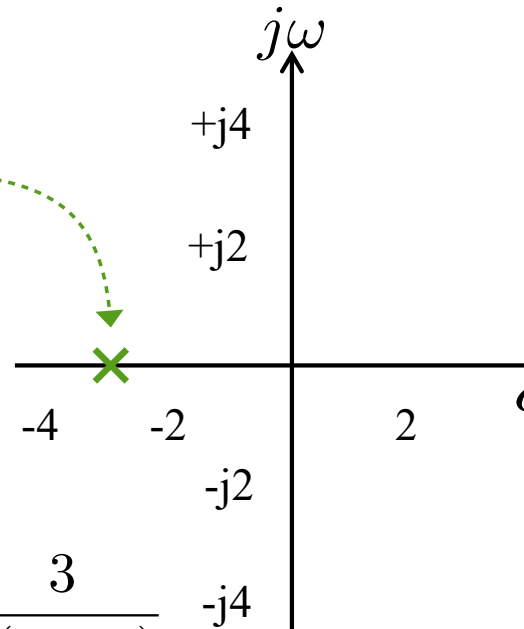
$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s + 3}$$

$$U(s) = \text{step} = \frac{1}{s}$$

$$Y(s) = U(s) \frac{1}{s + 3}$$

$$Y(s) = \frac{h_0}{s} K_{ss} \frac{\omega_b}{s + \omega_b} \rightarrow Y(s) = \frac{1}{3} \frac{3}{s(s + 3)}$$

$$Y(t) = h_0 K_{ss} (1 - e^{-\omega_b t}) \rightarrow y(t) = 0.33(1 - e^{-3t})$$

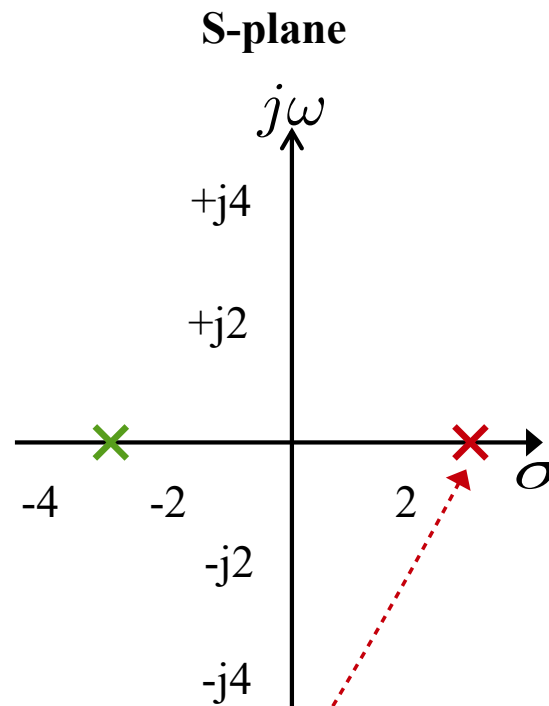


$$\tau = \frac{1}{3}$$

Poles (1st order): what is the consequence for $y(t)$?



(in the right half-plane)



When:

$$G_2(s) = \frac{Y_2(s)}{U(s)} = \frac{-1}{s-3}$$

$y_2(t)$?

Poles in the **right** half-plane

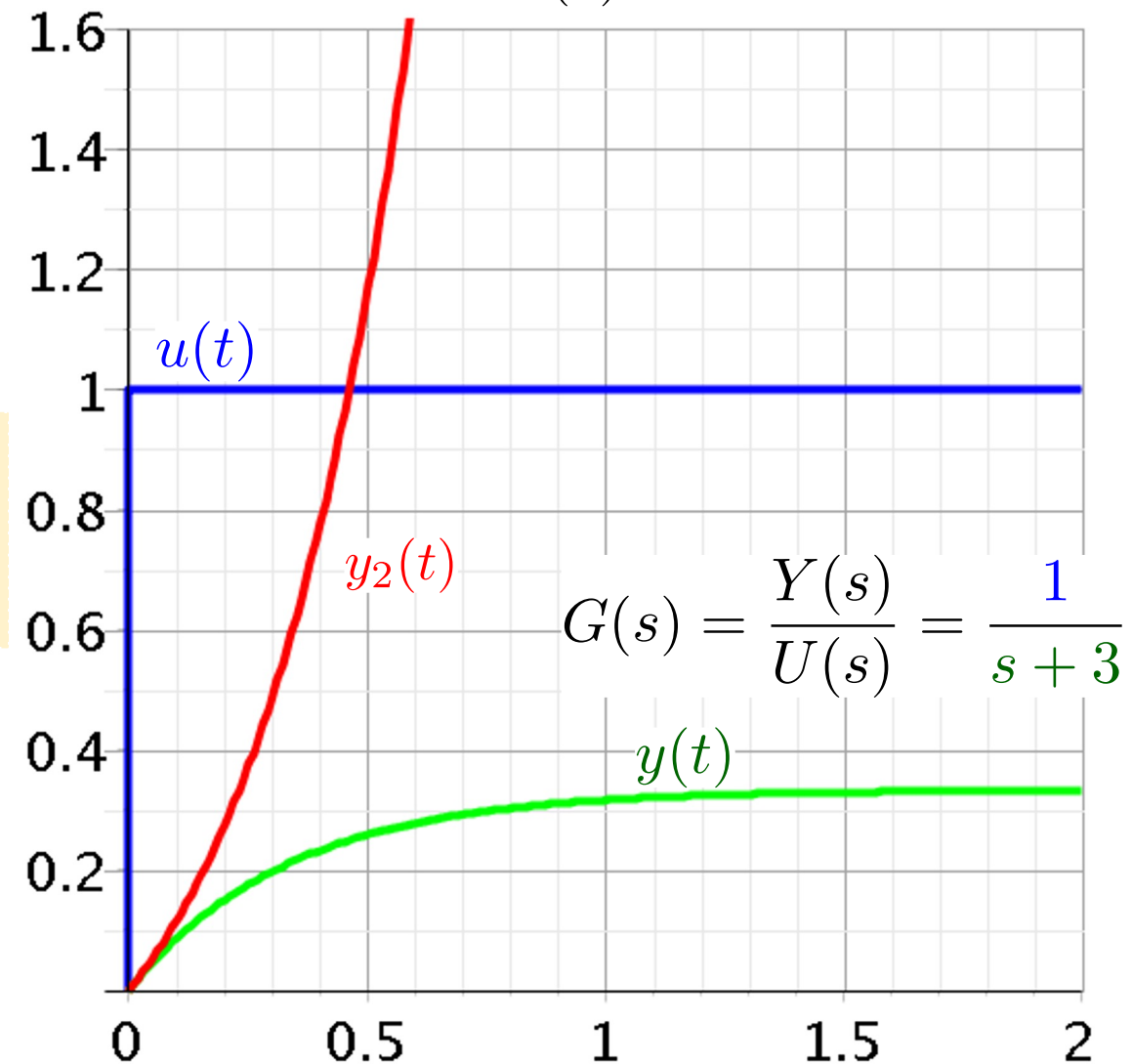


$$U(s) = \frac{1}{s}$$

$$y_2(t) = 0.33(1 - e^{+3t})$$

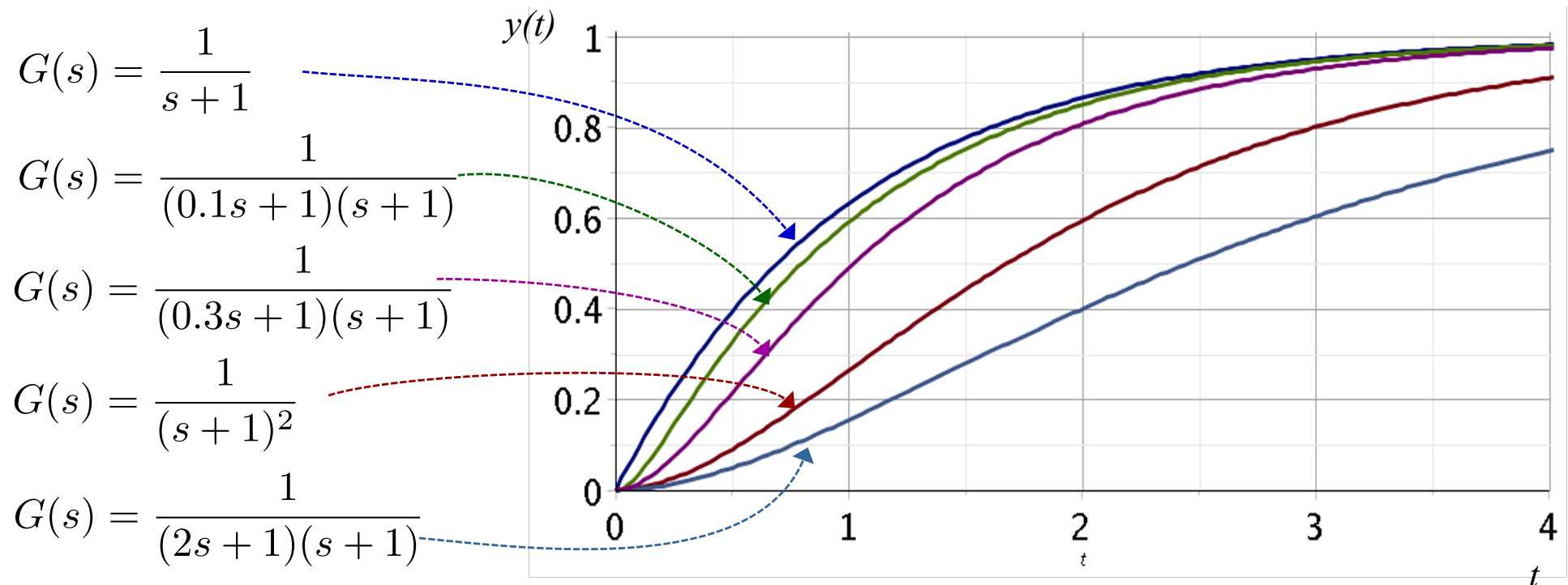
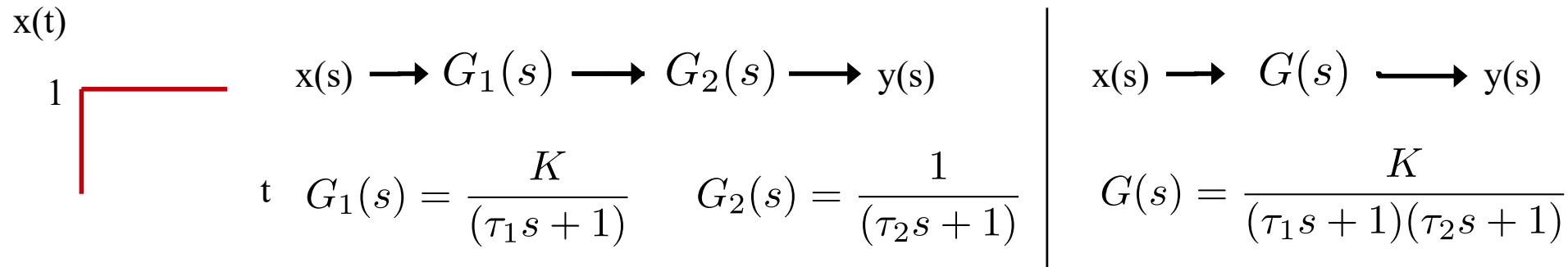
A pole in the right half-plane means that the transfer function is **unstable**.
Unstable: output never becomes bounded for stable input (except 0 input).

$$G_2(s) = \frac{Y_2(s)}{U(s)} = \frac{-1}{s - 3}$$



Real Poles (2nd order)

Step Response



Zeros (1st order)

What is the consequence for $y(t)$?

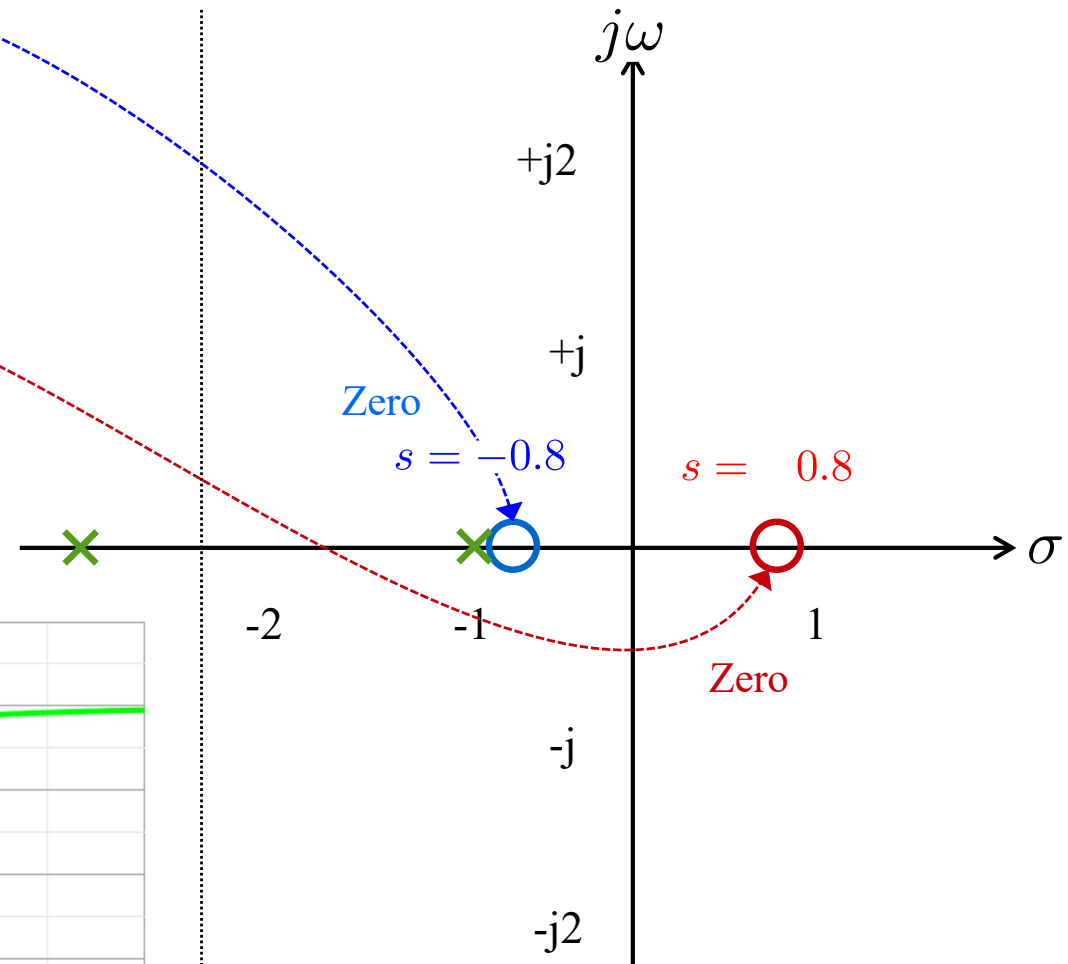
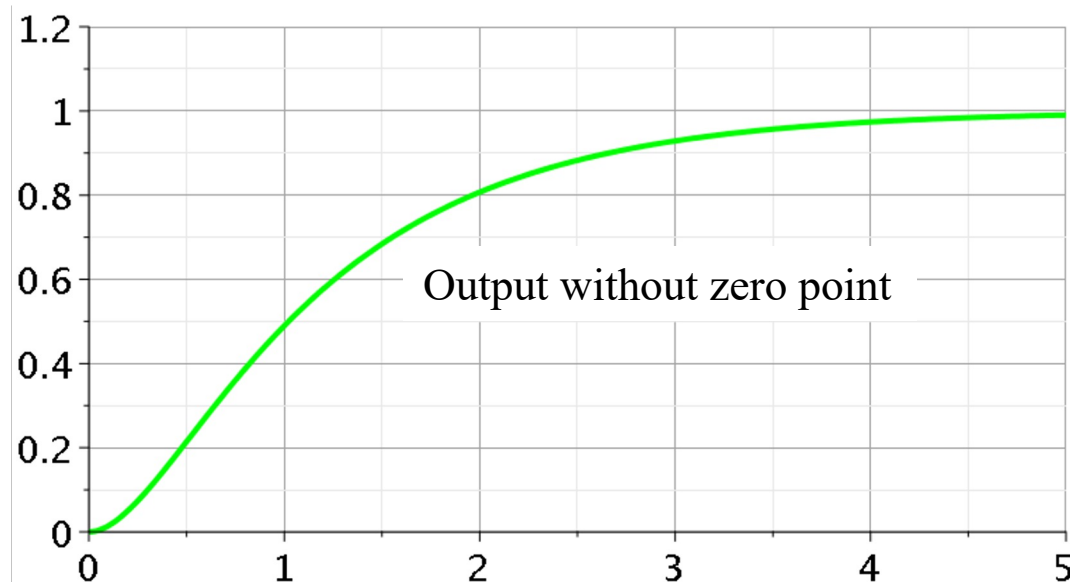
$$s = \sigma + j\omega$$



S-plane
(complex frequency plane)

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K (1.25s + 1)}{(0.3s + 1)(s + 1)}$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K (-1.25s + 1)}{(0.3s + 1)(s + 1)}$$



What does the zero point do?

Zero points (1st order - **Left half-plane**)

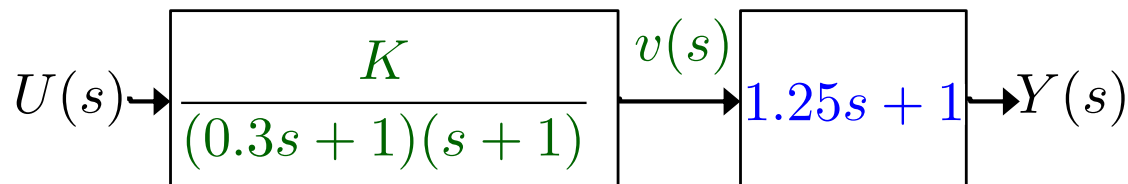
What is the consequence for $y(t)$?

$$s = \sigma + j\omega$$



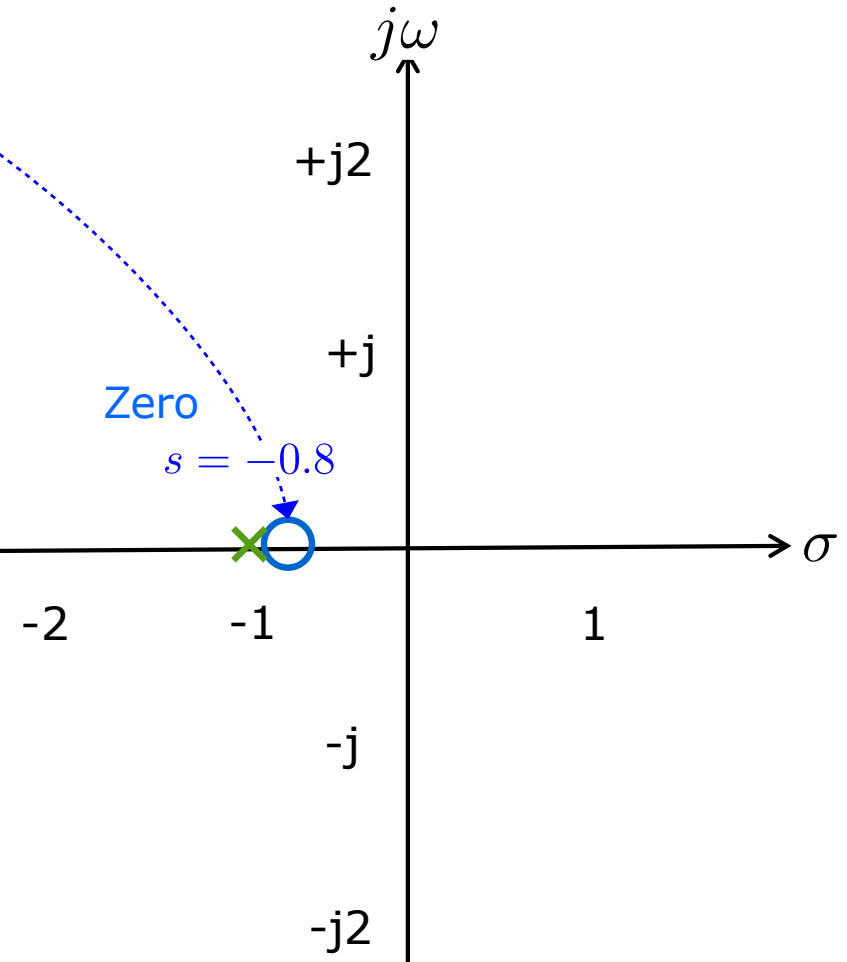
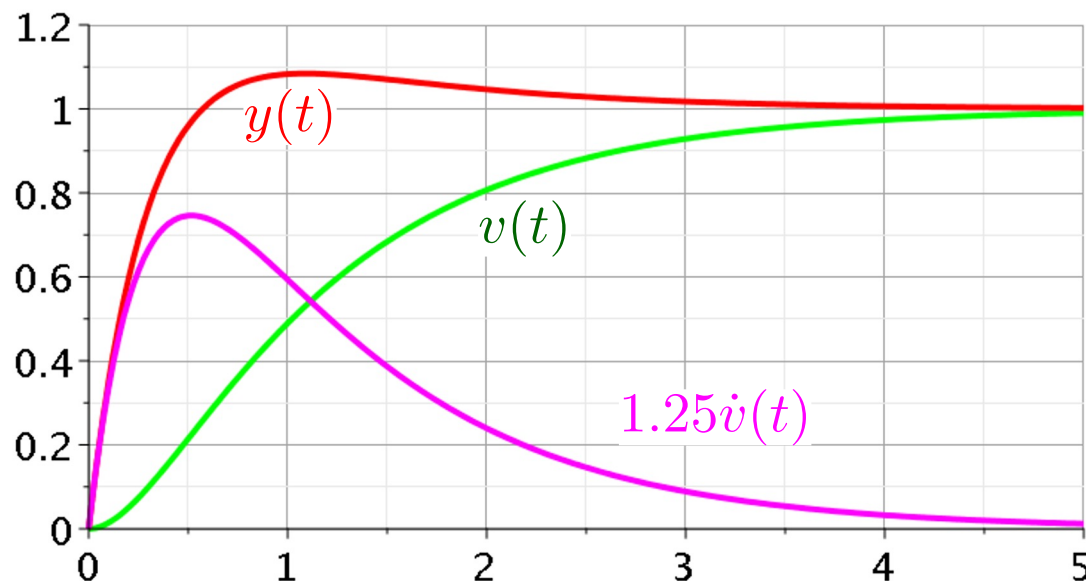
S-plane
(complex frequency plane)

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K (1.25s + 1)}{(0.3s + 1)(s + 1)}$$



$$Y(s) = 1.25sv(s) + v(s)$$

$$y(t) = 1.25\dot{v}(t) + v(t)$$

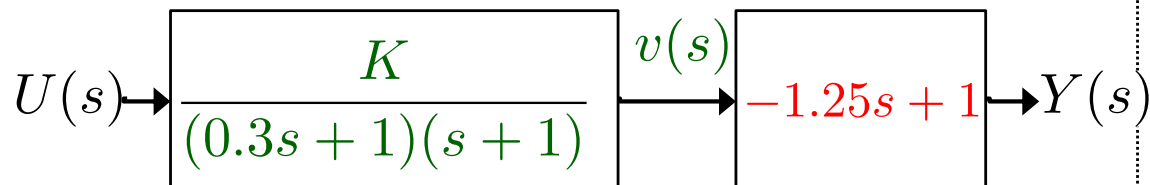


Zero points (1st order - **Right half-plane**)

What is the consequence for $y(t)$?

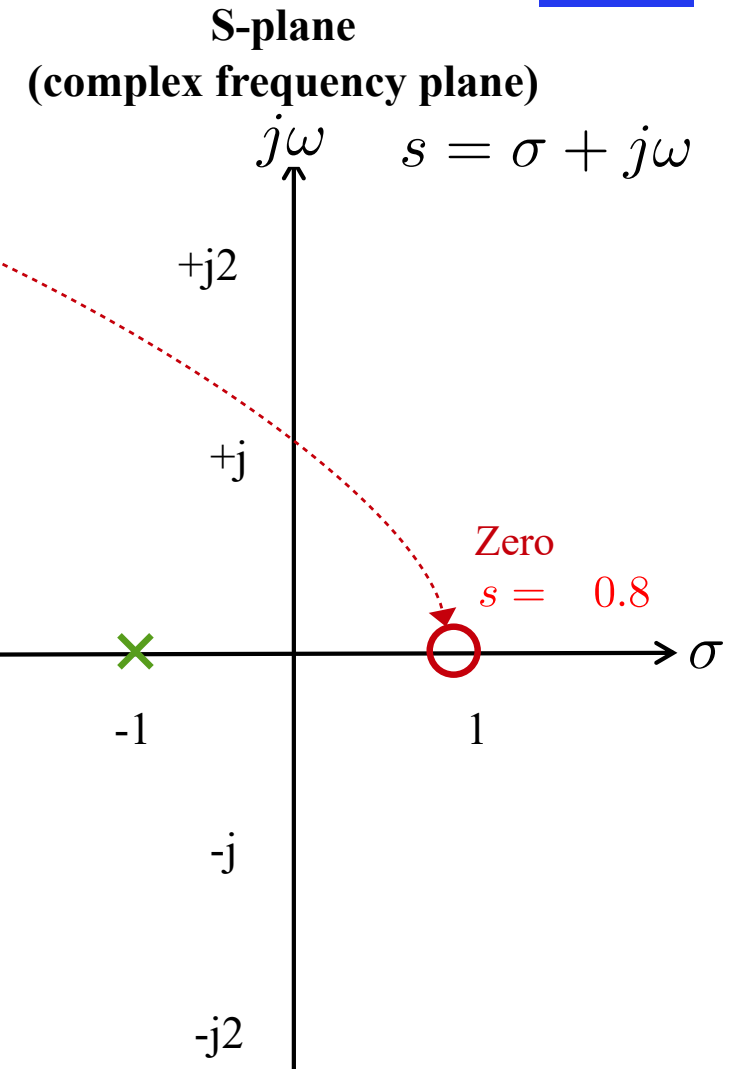
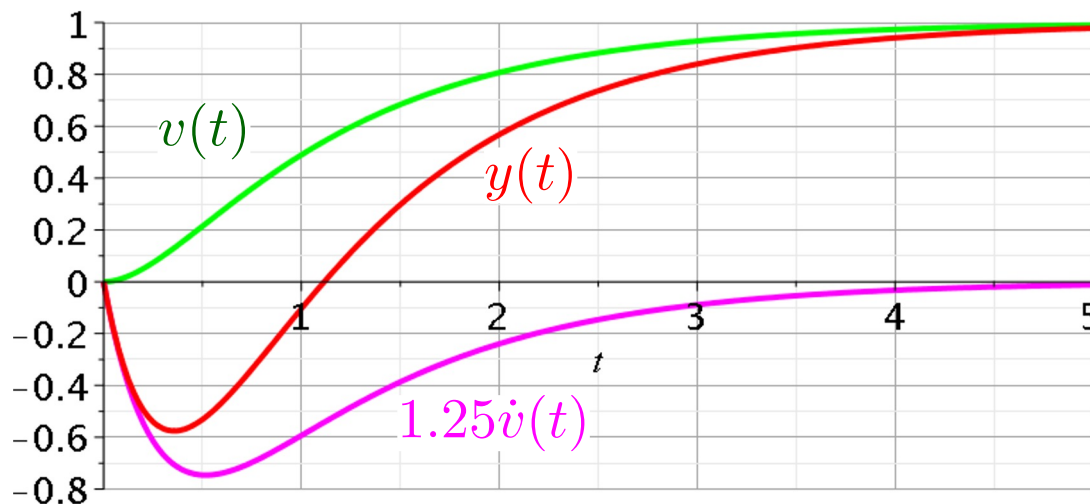


$$G(s) = \frac{Y(s)}{U(s)} = \frac{K (-1.25s + 1)}{(0.3s + 1)(s + 1)}$$



$$Y(s) = -1.25sv(s) + v(s)$$

$$y(t) = -1.25\dot{v}(t) + v(t)$$



Control question (S-plane)



1) Given the following transfer functions

$$G(s) = \frac{100}{s^2 + 7s + 10}$$

- a) Where does the system have poles and zeros?**
- b) Is the system stable?**

2) Given the following transfer

$$G(s) = \frac{s - 2}{0.5s - 2}$$

- a) What is steady-state gain?**
- b) Where does the system have poles and zeros?**
- c) Is the system stable?**

Solution 1)



1) Given the following transfer function

$$G(s) = \frac{100}{s^2 + 7s + 10}$$

a) Where does the system have poles and zeros?

$$s^2 + 7s + 10 = 0$$

$$s = \frac{-7 \pm \sqrt{49 - 4 \cdot 10}}{2}$$

$$\text{Poles} \quad s = -5$$
$$s = -2$$

b) Is the system stable?

YES. Poles are in the left half-plane

Roots of denominator:

$$Ax^2 + Bx + C = 0$$

$$x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

Solution 2)

2) Given the following transfer function

$$G(s) = \frac{s - 2}{0.5s - 2}$$

a) What is the steady-state gain?

$$K_{ss} = \lim_{s \rightarrow 0} G(s) \quad K_{ss} = 1$$

b) Where do the system have poles and zeros?

$$\text{Pole } s = \frac{2}{0.5} = +4$$

$$\text{Zero: } s = +2$$

c) Is the system stable?

No, there is a pole in the right half-plane

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2nd order system

Complex Poles



$$G(s) = \frac{b_0}{s^2 + a_1 s + a_0} \Big|_{a_1^2 < 4a_0}$$

→ complex roots

Roots of denominator:

$$s^2 + a_1 s + a_0$$

$$s_{1,2} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0}}{2}$$

$$a_1^2 - 4a_0 \geq 0 \text{ real}$$

$$a_1^2 - 4a_0 < 0 \text{ complex}$$

2nd order system

Complex Poles



$$G(s) = K_{ss} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n = natural frequency [rad/s]

ζ (zeta) = damping factor

$$\zeta < 1 \Rightarrow (2\zeta\omega_n)^2 < 4\omega_n^2$$

→ complex roots

Filter technology

$$G(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

Q = Quality factor

$$\zeta = \frac{1}{2Q}$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

1. Natural frequency ω_n

The frequency of **free** oscillation if there is no damping at all.

2. Damping frequency ω_d

The frequency of **free** oscillation when there is damping with a damping ratio $0 < \zeta < 1$.

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

3. Resonance frequency ω_r

The frequency of **oscillatory input** for which the output amplitude is maximized. It only occurs when $0 < \zeta < 1$.

$$\left. \frac{\partial |G(\omega)|}{\partial \omega} \right|_{\omega=\omega_r} = 0 \Rightarrow \omega_r = \omega_n \sqrt{1 - 2\zeta^2}$$

Peak magnitude

$$M_P = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$$

2nd order system - Step Response



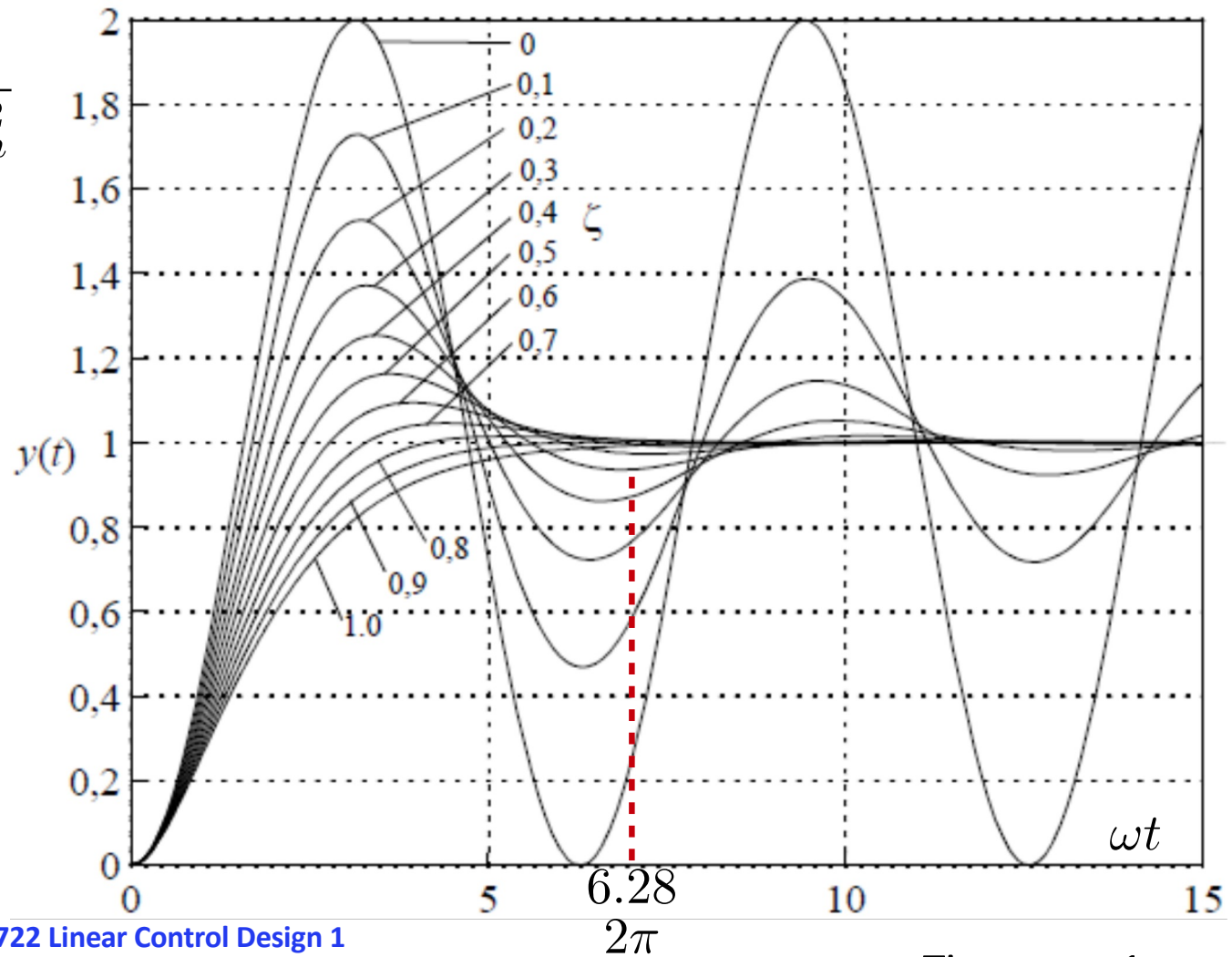
$$X(s) \longrightarrow G(s) \longrightarrow Y(s)$$

Unit step response

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ζ (zeta) = damping factor

ω_n = natural frequency [rad/s]



Control points

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- $\zeta < 0 \rightarrow$ instability
- $\zeta = 0 \rightarrow$ non-decaying oscillation (undamped)
- $0 < \zeta < 1 \rightarrow$ damped oscillation (damped / Underdamped)
- $\zeta = 1 \rightarrow$ no oscillation (critically damped)
- $\zeta > 1 \rightarrow$ no oscillation (overdamped)

Complex poles

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



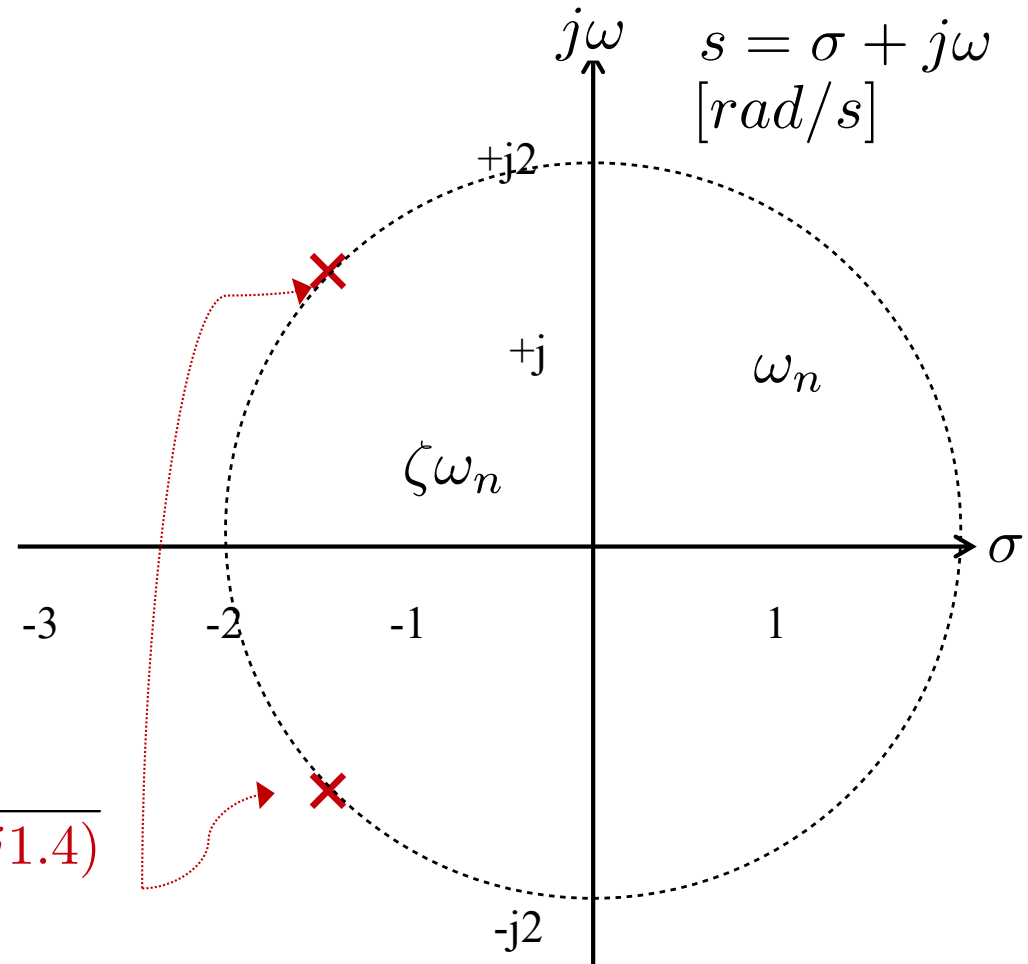
S-plane
(complex frequency plane)

$$G(s) = \frac{4}{(s^2 + 2 \cdot 0.707 \cdot 2s + 4)}$$

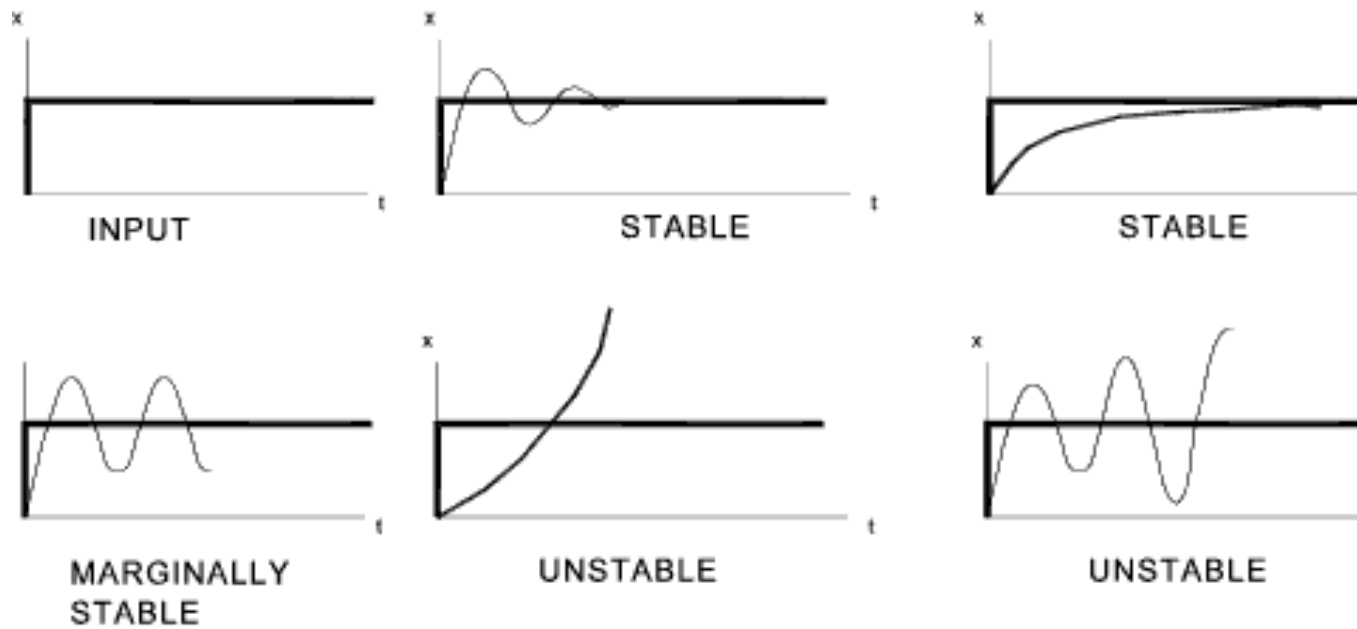
$$\zeta = 0.707$$

$$\omega_n = 2 \text{ [rad/s]}$$

$$G(s) = \frac{4}{(s + 1.4 + j1.4)(s + 1.4 - j1.4)}$$



(2nd order poles always come in complexly conjugated pairs)



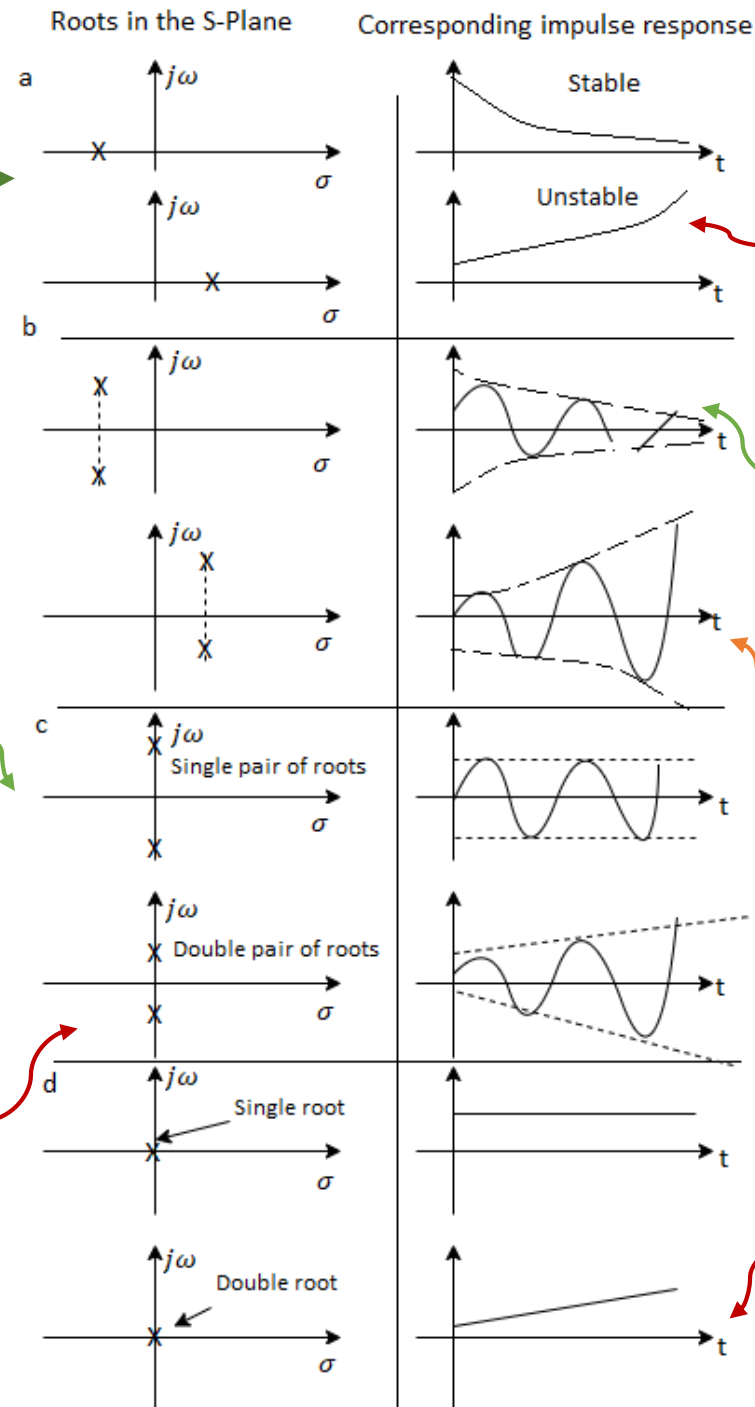
Pole locations

Output of the system with pole / pole pairs as shown with an impulse as input.

Poles on the LEFT half-plane (LHP): **STABILITY**

Single Pair of poles on the imaginary axis: **marginally stable**

Multiple Pairs of poles on the imaginary axis: **unstable**



Unstable systems

- Small perturbations remove system from equilibrium (e.g., inverted pendulum)
- Bounded input gives unbounded output (e.g., constant motor torque $\rightarrow$ infinite angular position)
- If there is at least one pole in the **RHP**

$$G_1(s) = \frac{1}{s-1} \Leftrightarrow \dot{y} = y + u(t) \Rightarrow y(t) = y_0 e^t + \int_0^t e^{t-\tau} u(\tau) d\tau \quad \rightarrow \quad \boxed{\lim_{t \rightarrow \infty} |y(t)| = \infty}$$

- **Multiple** poles on the imaginary axis (on top of each other)

$$G_2(s) = \frac{1}{(s^2 + 1)^2} \xrightarrow{u(s)=1} y^{(4)} + \ddot{y} + y = 1\delta(t) \Rightarrow y(t) = 0.5 \sin(t) - 0.5t \cos(t)$$

- How can we stabilize an open-loop unstable system? **WAIT FOR LECTURE 10!**

Control Questions



1) Two systems have the following transfer functions

$$G_1(s) = \frac{10}{s + 100} \qquad G_2(s) = \frac{1000}{s^2 + 14s + 100}$$

Which system is the fastest?

2) Does this system $G_2(s) = \frac{1000}{s^2 + 10s + 100}$ have

- a) Complex poles?
- b) What is the break frequency?
- c) What is the damping factor?
- d) What is the overshoot in % when the input is a unit step?

Control Questions - **Solution**

1) Two systems have the following transfer functions:

$$G_1(s) = \frac{10}{s + 100} \qquad G_2(s) = \frac{1000}{s^2 + 14s + 100}$$

Which is the fastest system?

System G1, real pole $s = -100$

System G2, complex poles

$$s = \frac{-14 \pm \sqrt{14^2 - 4 \cdot 100}}{2}$$

$$s = -7 \pm 7.14j$$

So, system **G1** is the fastest one

Control Questions - Solution



$$G_2(s) = \frac{1000}{s^2 + 10s + 100}$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$x = \frac{B \pm \sqrt{B^2 - 4AC}}{2A}$$

2)

a) Does the system have complex poles?

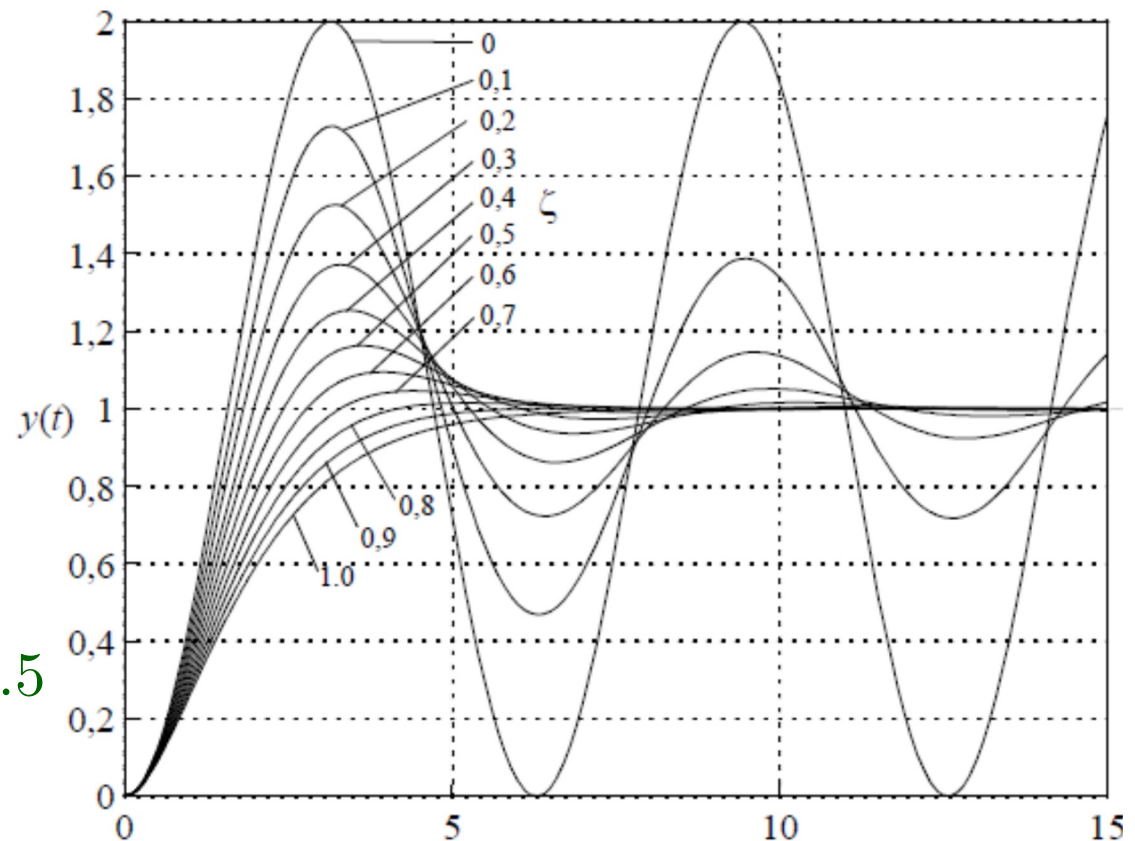
yes $\Rightarrow s = -7 \pm 8.6j$

b) what is the natural frequency?

$$\omega_n = \sqrt{100} = 10$$

c) what is the damping factor?

$$2\zeta\omega_n = 10 \Rightarrow \zeta = \frac{10}{2\omega_n} \Rightarrow \zeta = 0.5$$



d) What is the overshoot in % when the input is a unit step?

Based on graph: 17%

Example

For a system, the following transfer function has been calculated:

$$G(s) = \frac{s^2 + 7s + 324}{(0.01s + 1)(s^2 + 5s + 400)}$$

- a) Are there real poles and zeros?
- b) Where are the complex poles and zeros?
- c) Is the system stable?
- d) Draw the poles and zeros in the s-plane
- e) What are the resonance frequency and damping factor?
- f) Which is the most dominant pole?

Example - Solution

For a system, the following transfer function has been calculated:

$$G(s) = \frac{s^2 + 7s + 324}{(0.01s + 1)(s^2 + 5s + 400)}$$

a) Are there real poles and zeros?

Real pole: $s = -100$ (rad/sec)

Example - Solution

For a system, the following transfer function has been calculated:

$$G(s) = \frac{s^2 + 7s + 324}{(0.01s + 1)(s^2 + 5s + 400)}$$

b) Where are the complex poles and zeros?

$$a_1^2 < 4a_0 \quad (a_2 = 1)$$

$$(a_1^2 = 25) < (4a_0 = 1600) \Rightarrow \text{complex poles}$$

$$(a_1^2 = 49) < (4a_0 = 1296) \Rightarrow \text{complex zeros}$$

Roots of the denominator:

$$s^2 + a_1s + a_0$$

$$s_{1,2} = \frac{a_1 \pm \sqrt{a_1^2 - 4a_0}}{2}$$

$$a_1^2 - 4a_0 \geq 0 \text{ real roots}$$

$$a_1^2 - 4a_0 < 0 \text{ complex roots}$$

Example - Solution

b) Where are the complex poles and zeros?

Complex zeros:

$$s^2 + 7s + 324 = 0 \Rightarrow s = \frac{-7}{2} \pm \frac{\sqrt{7^2 - 4 \cdot 324}}{2}$$

$$s = -3.5 \pm 17.7j$$

Complex poles:

$$s^2 + 5s + 400 = 0 \Rightarrow s = \frac{-5}{2} \pm \frac{\sqrt{5^2 - 4 \cdot 400}}{2}$$

$$s = -2.5 \pm 19.8j$$

MATLAB:

```
G1 = tf([1 7 324],[1 5 400])
```

```
G2 = tf([1],[0.01 1])
```

```
pole(G1*G2)
```

```
ans =
```

```
1.0e+02 *
```

```
-1.0000 + 0.0000i
```

```
-0.0250 + 0.1984i
```

```
-0.0250 - 0.1984i
```

```
zero(G1*G2)
```

```
ans =
```

```
-3.5000 +17.6564i
```

```
-3.5000 -17.6564i
```

Example - Solution

$$G(s) = \frac{s^2 + 7s + 324}{(0.01s + 1)(s^2 + 5s + 400)}$$

MATLAB

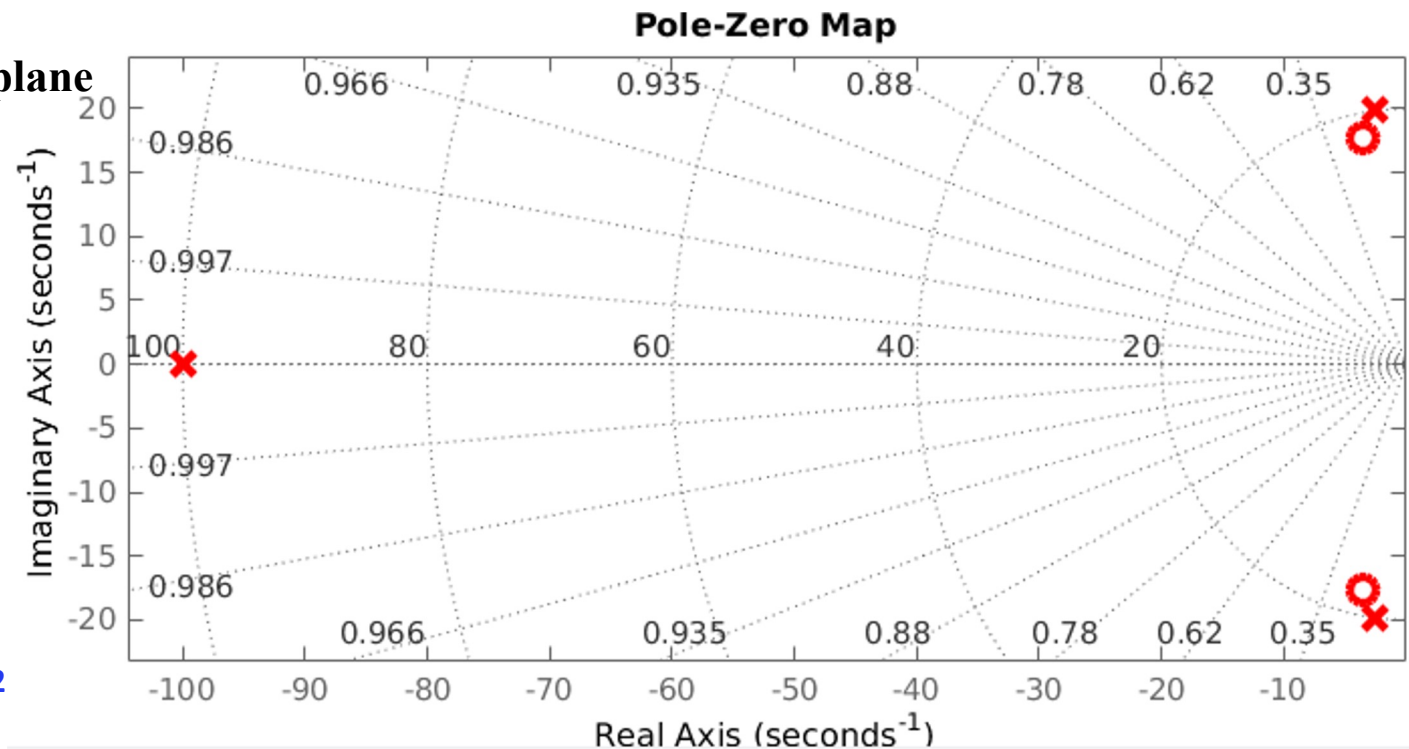
c) Is the system stable?

Poles: $s = -100$
 $s = -2.5 \pm 19.8j$

All in the left half-plane → **stable**.

figure(31)
 pzplot(G1*G2)
 grid on
 axis equal

d) Draw poles and zeros in s-plane



Example - Solution

$$G(s) = \frac{s^2 + 7s + 324}{(0.01s + 1)(s^2 + 5s + 400)}$$

e) Natural frequency and damping factor?

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Natural frequency

Pole: $\omega_{np} = \sqrt{400} = 20 \text{ rad/sec}$

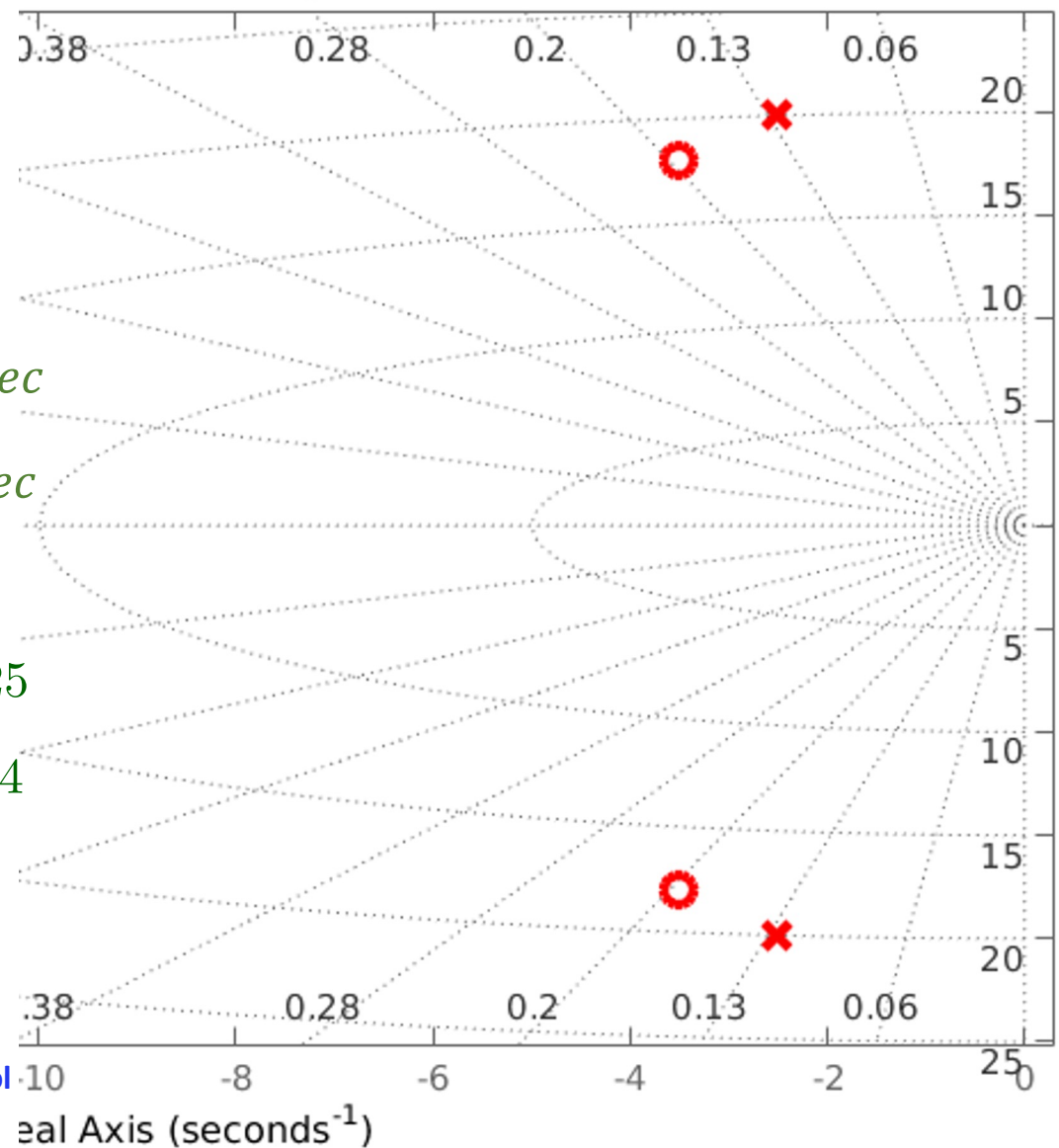
Zero: $\omega_{nz} = \sqrt{324} = 18 \text{ rad/sec}$

Damping factor

Pole: $2\zeta_p\omega_{np} = 5 \Rightarrow \zeta_p = 0.125$

Zero point: $2\zeta_z\omega_{nz} = 7 \Rightarrow \zeta_z = 0.194$

Pole-Zero Map



Example - Solution

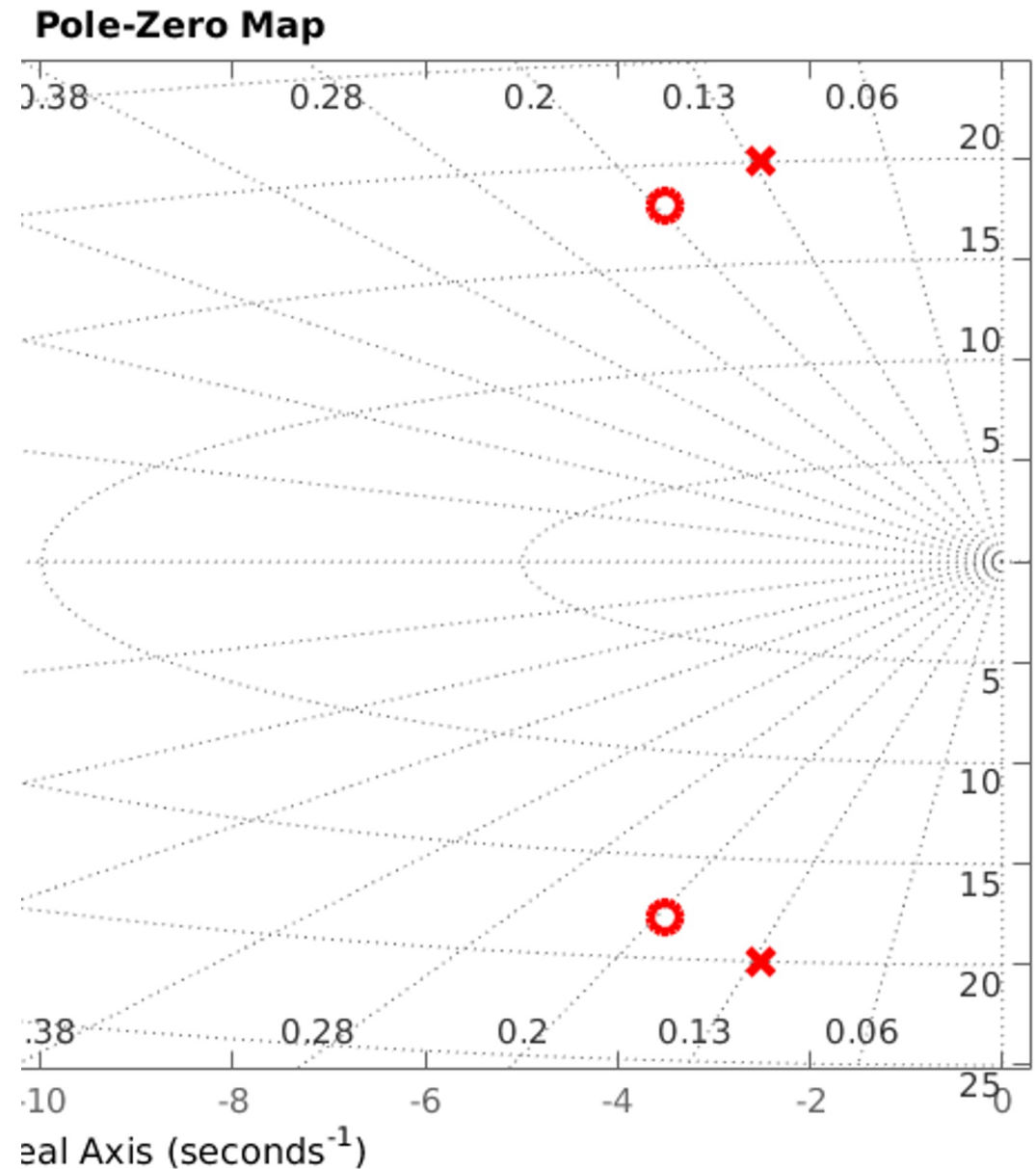
$$G(s) = \frac{s^2 + 7s + 324}{(0.01s + 1)(s^2 + 5s + 400)}$$



f) Most dominant pole?

The one with the smallest (negative) frequency $\rightarrow s = 20$

What if the zero points were on top of poles?
(zero-pole cancellation)



Example - Solution



$$G(s) = \frac{s^2 + 7s + 324}{(0.01s + 1)(s^2 + 5s + 400)}$$

If the input is a unit step, then what is (output):

g) Rise time?

h) Overshoot in %?

MATLAB

```
stepinfo(G1*G2)
```

```
ans =
```

RiseTime: 0.0119

SettlingTime: 1.0129

SettlingMin: 0.8266

SettlingMax: 1.2371

Overshoot: 23.7148

Undershoot: 0

Peak: 1.2371

PeakTime: 0.0461

